

LAP EXPEIMENT-20/07/2026

Set 1: Monthly Sales Data

Dataset: Monthly Sales

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
library(ggplot2)

sales <- data.frame(

  Month = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),

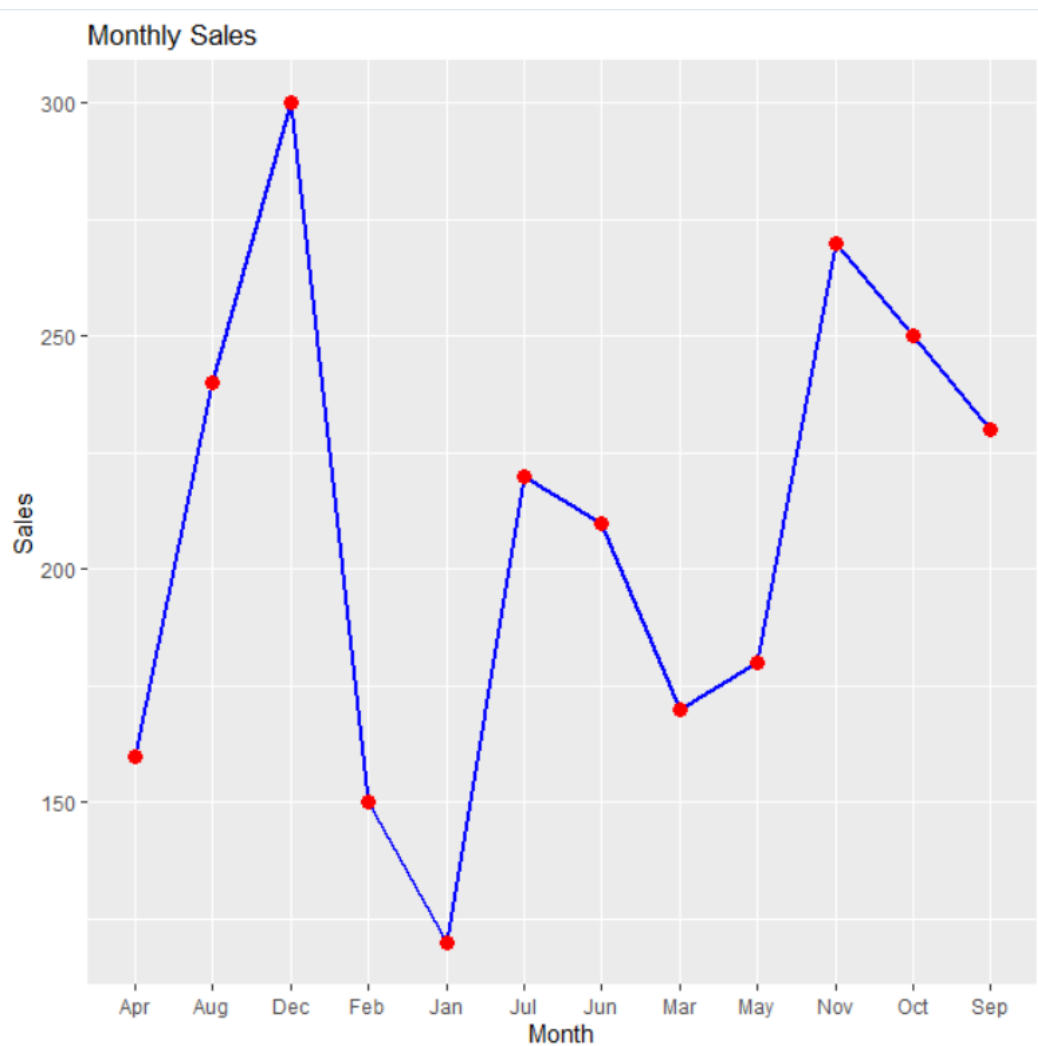
  Sales = c(120, 150, 170, 160, 180, 210, 220, 240, 230, 250, 270, 300),

  Advertising = c(20, 25, 30, 28, 35, 40, 45, 50, 48, 55, 60, 65))

ggplot(sales, aes(x = Month, y = Sales, group = 1)) +

  geom_line(color = "blue", linewidth = 1) +

  geom_point(color = "red", size = 3) + labs(title = "Monthly Sales", x = "Month", y = "Sales")
```

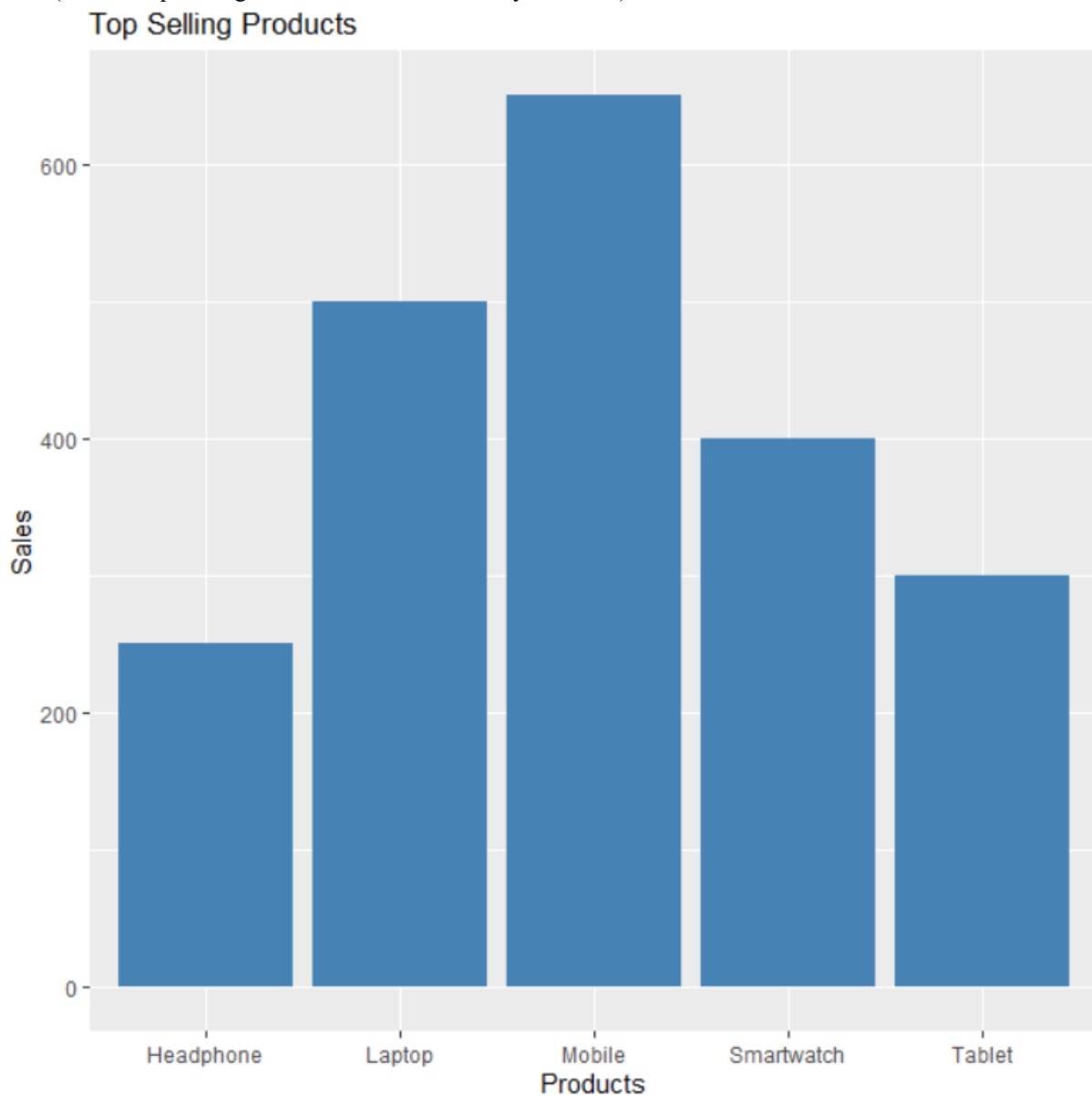


2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

```
library(ggplot2)

products <- data.frame(
  Product = c("Laptop", "Mobile", "Tablet", "Headphone", "Smartwatch"),
  Sales = c(500, 650, 300, 250, 400))

ggplot(products, aes(x = Product, y = Sales)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Top Selling Products", x = "Products", y = "Sales")
```



3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

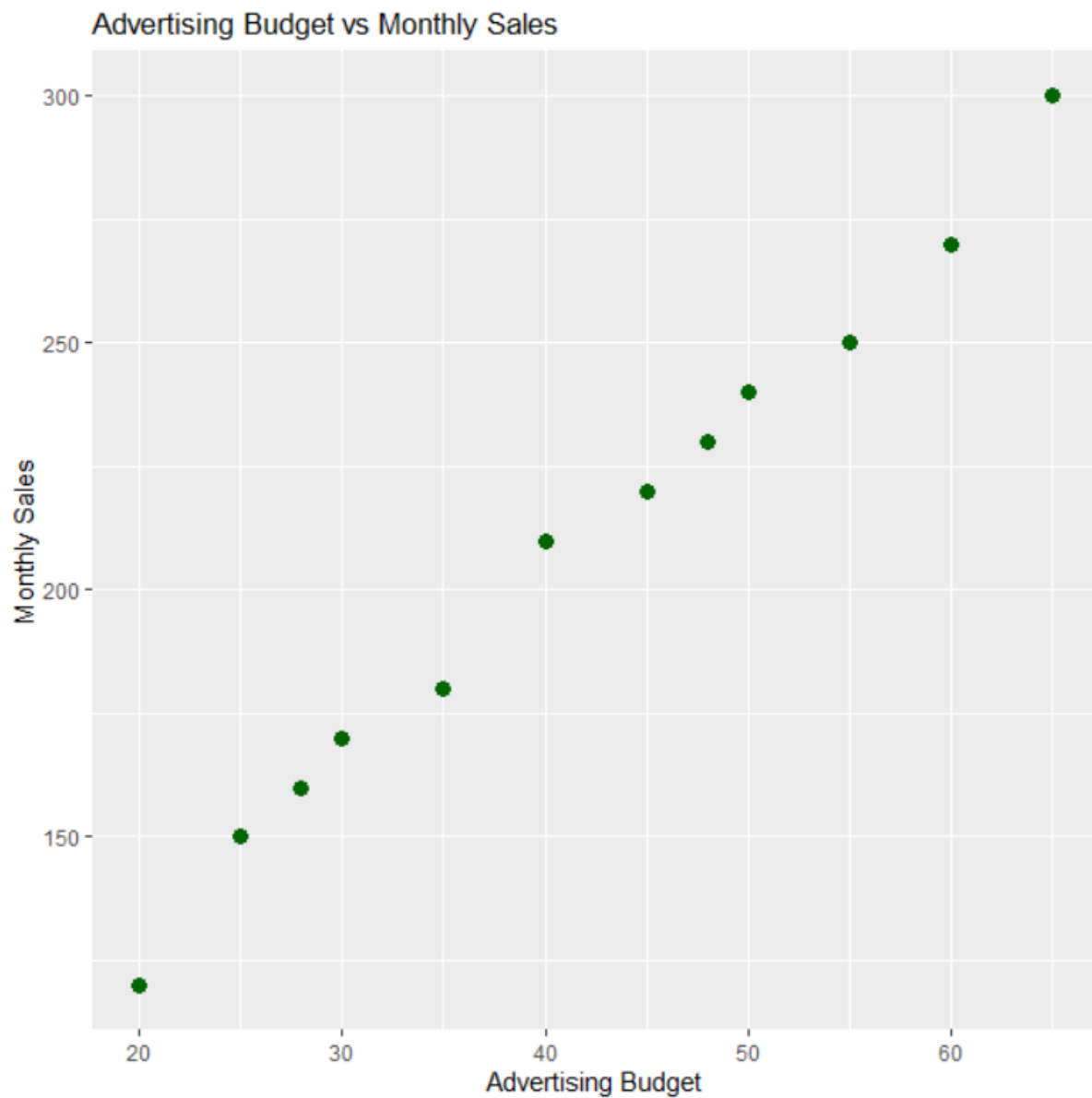
Explain the insights drawn from the scatter plot.

```
library(ggplot2)
```

```
ggplot(sales, aes(x = Advertising, y = Sales)) +
```

```
  geom_point(color = "darkgreen", size = 3) +
```

```
  labs( title = "Advertising Budget vs Monthly Sales", x = "Advertising Budget", y = "Monthly Sales")
```



4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

```
library(ggplot2)
library(gridExtra)

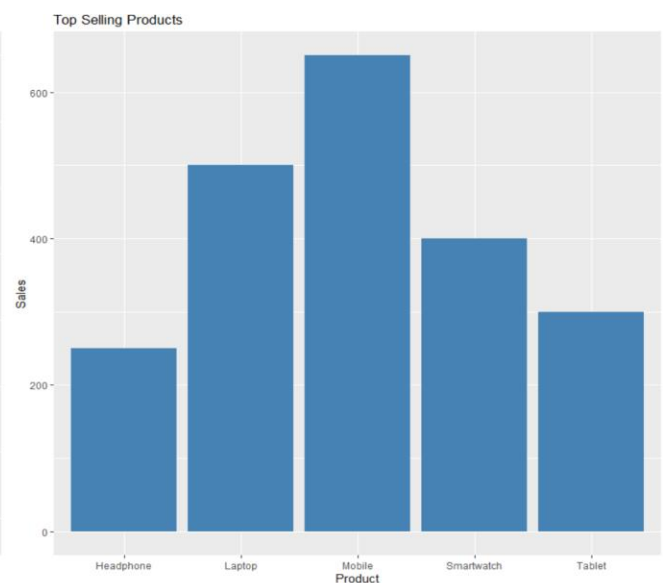
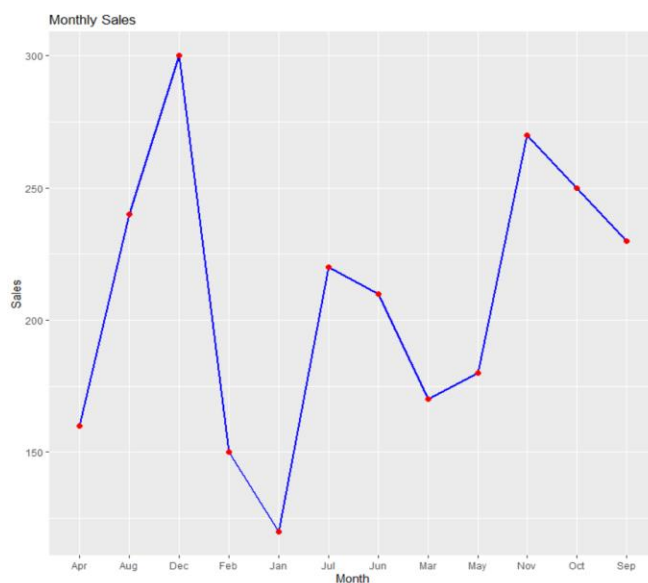
sales <- data.frame(
  Month = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
  Sales = c(120, 150, 170, 160, 180, 210, 220, 240, 230, 250, 270, 300))

products <- data.frame(
  Product = c("Laptop", "Mobile", "Tablet", "Headphone", "Smartwatch"),
  Sales = c(500, 650, 300, 250, 400))

p1 <- ggplot(sales, aes(x = Month, y = Sales, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 2) +
  labs(title = "Monthly Sales", x = "Month", y = "Sales")

p2 <- ggplot(products, aes(x = Product, y = Sales)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Top Selling Products",
       x = "Product",
       y = "Sales")

gridExtra::grid.arrange(p1, p2, ncol = 2)
```



Set 2

Month	Sales (in \$)
January	15000
February	18000
March	22000
April	20000
May	23000

Customer Feedback Analysis
Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the Chart.

```
library(ggplot2)
```

```
customer <- data.frame(
```

```
  CustomerID = c(1, 2, 3, 4, 5),
```

```
  Age = c(25, 30, 35, 28, 40),
```

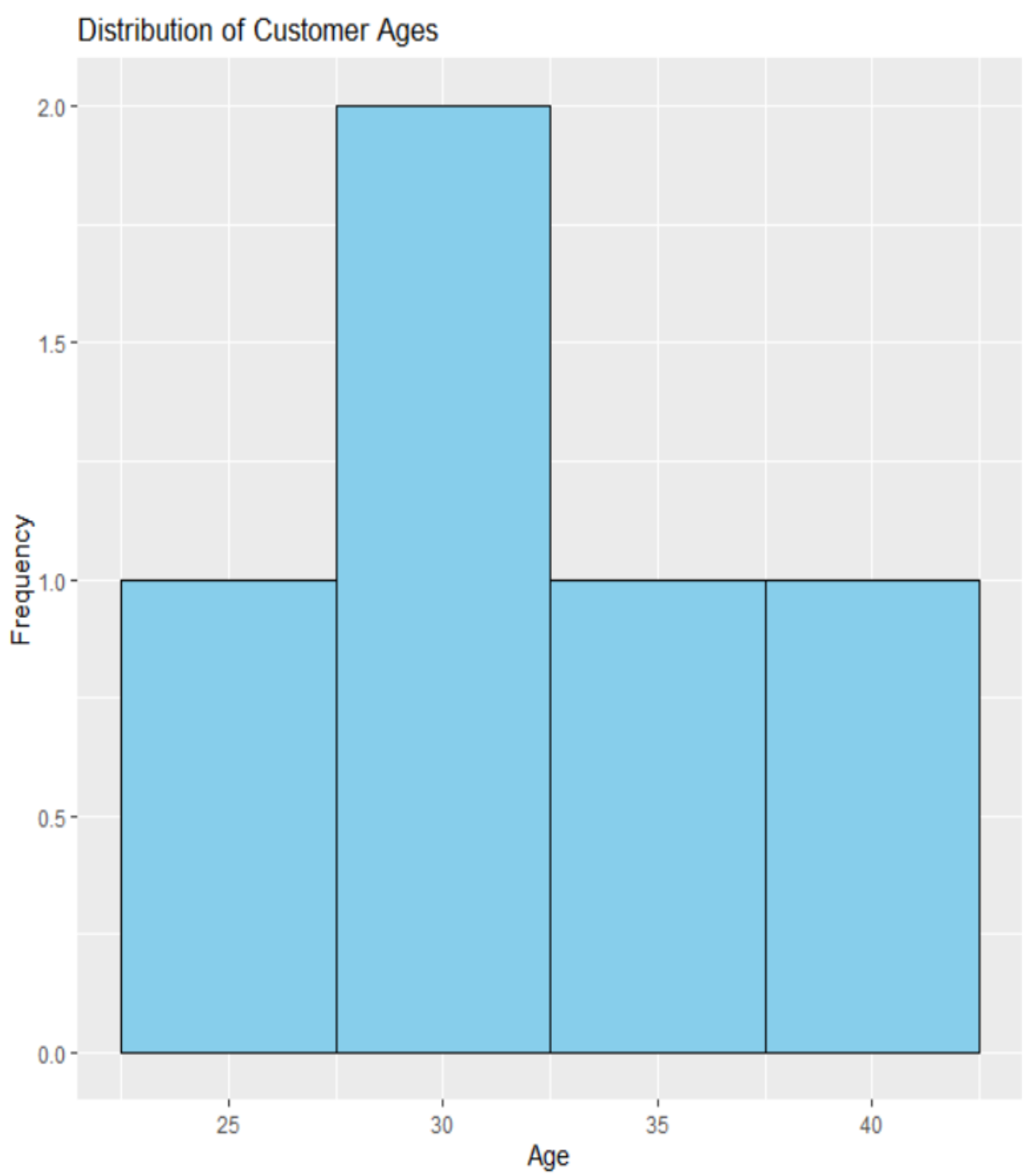
```
  Satisfaction = c(4, 5, 3, 4, 5),
```

```
  stringsAsFactors = FALSE) print(customer)
```

```
ggplot(customer, aes(x = Age)) +
```

```
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
```

```
  labs(title = "Distribution of Customer Ages", x = "Age", y = "Frequency")
```



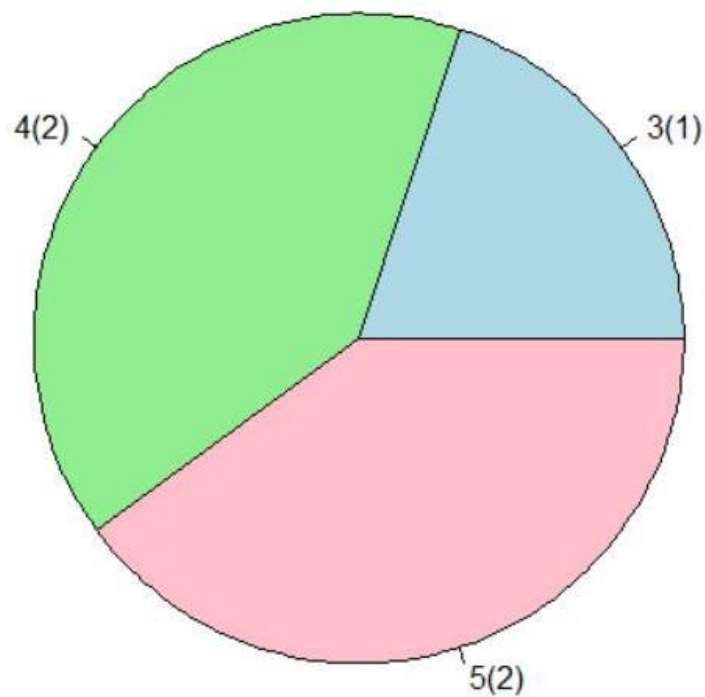
2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

```
library(ggplot2)

customer <- data.frame(
  CustomerID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction = c(4,5,3,4,5))

score <- table(customer$Satisfaction)
pie(score,
  labels = paste(names(score), "(", score, ")", sep=""),
  main = "Distribution of Customer Satisfaction Scores",
  col = c("lightblue", "lightgreen", "pink"))
```

Distribution of Customer Satisfaction Scores



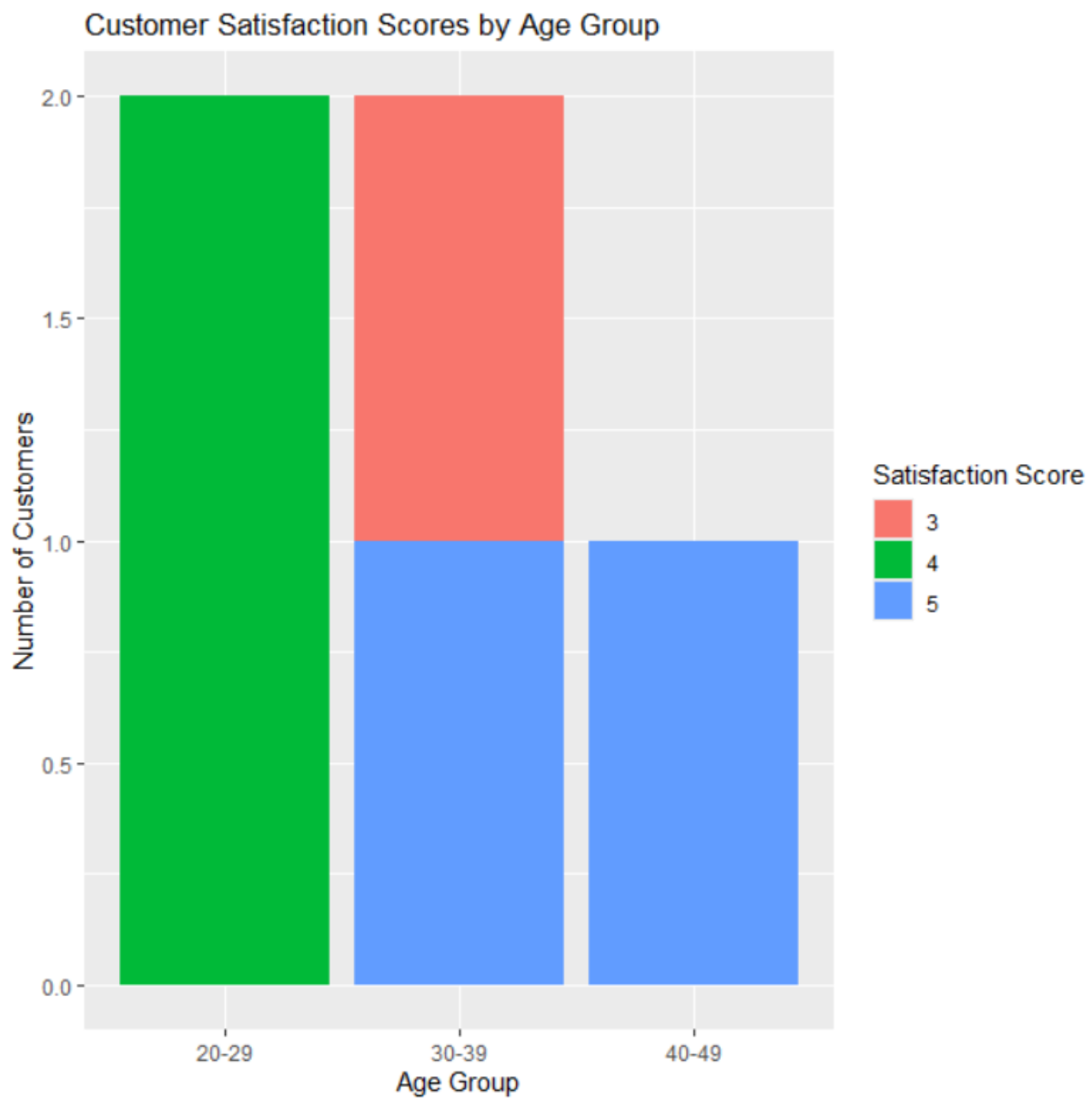
3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
library(ggplot2)

customer <- data.frame(
  CustomerID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction = c(4,5,3,4,5))

customer$AgeGroup <- c("20-29","30-39","30-39","20-29","40-49")

ggplot(customer, aes(x = AgeGroup, fill = factor(Satisfaction))) + geom_bar() +
  labs( title = "Customer Satisfaction Scores by Age Group", x = "Age Group",
    y = "Number of Customers", fill = "Satisfaction Score")
```



4. Develop a word cloud from open-ended customer feedback to identify prevalent customer.

```
library(ggplot2)
```

```
feedback <- c(
```

```
  "Excellent", "Excellent", "Good", "Service",
```

```
  "Support", "Product", "Quality", "Delivery",
```

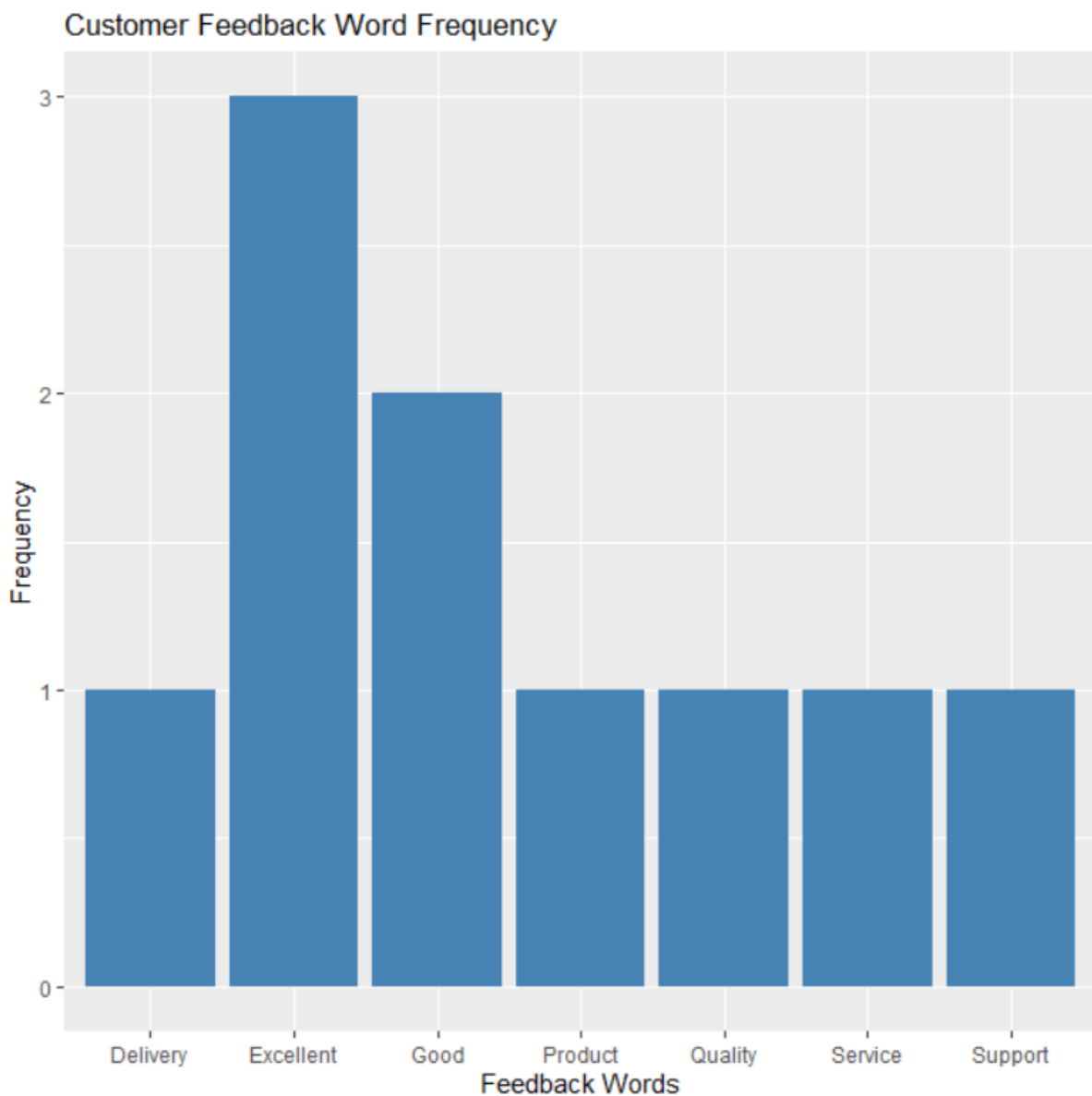
```
  "Good", "Excellent")
```

```
freq <- as.data.frame(table(feedback))
```

```
ggplot(freq, aes(x = feedback, y = Freq)) +
```

```
geom_bar(stat = "identity", fill = "steelblue") +
```

```
labs(title = "Customer Feedback Word Frequency", x = "Feedback Words", y = "Frequency")
```



Set 3: Employee Performance Evaluation

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.
4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

#CODE

Load library

```
library(ggplot2)
```

Employee Performance Dataset

```
employee <- data.frame(  
  EmployeeID = c(1,2,3,4,5),  
  Department = c("Sales","HR","Marketing","Sales","HR"),  
  YearsService = c(5,3,7,4,2),  
  PerformanceScore = c(85,92,78,90,76)  
)
```

1. Line Chart - Performance Trend

```
ggplot(employee, aes(x = YearsService, y = PerformanceScore, group = 1)) +  
  geom_line(color = "blue", size = 1) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Employee Performance Trend",  
    x = "Years of Service",  
    y = "Performance Score"  
  )
```

2. Bar Chart - Employees by Department

```
ggplot(employee, aes(x = Department, fill = Department)) +  
geom_bar() +  
  labs(  
    title = "Employees by Department",  
    x = "Department",  
    y = "Number of Employees"  
  )
```

3. Scatter Plot - Years of Service vs Performance

```
ggplot(employee, aes(x = YearsService, y = PerformanceScore, color = Department)) +  
  geom_point(size = 4) +  
  labs(  
    title = "Years of Service vs Performance Score",  
    x = "Years of Service",  
    y = "Performance Score"  
  )
```

4. Dashboard (Display Line Chart and Bar Chart Together)

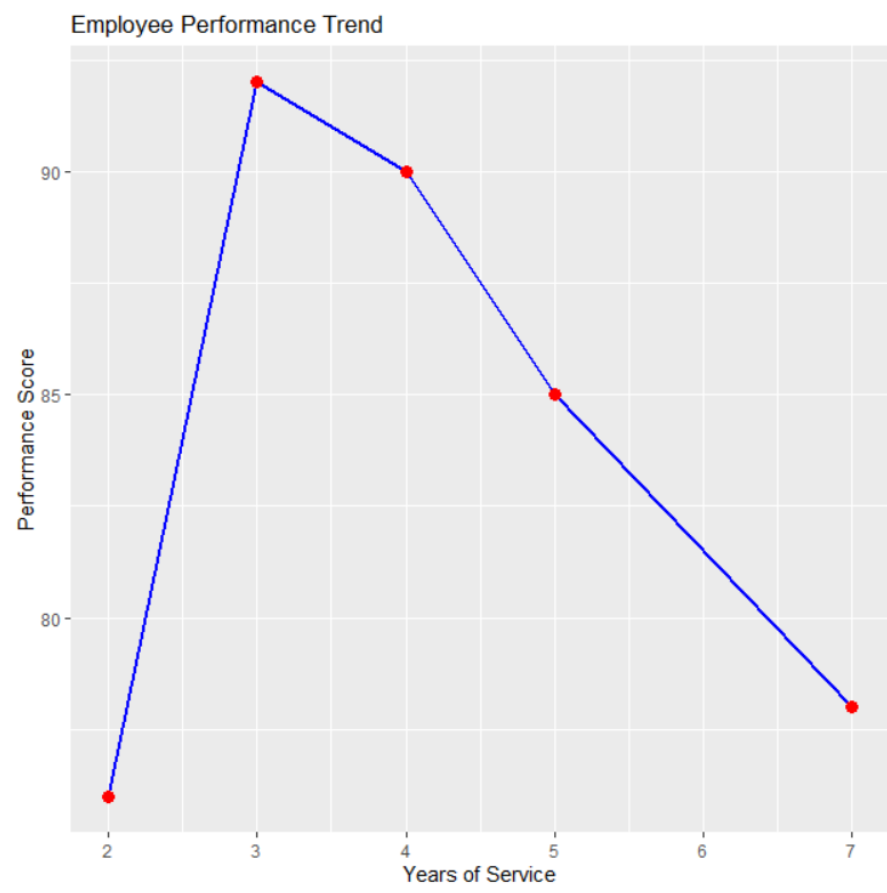
```
library(gridExtra)
```

```
p1 <- ggplot(employee, aes(x = YearsService, y = PerformanceScore, group = 1)) +  
  geom_line(color = "blue", size = 1) +  
  geom_point(color = "red", size = 3) +  
  labs(title = "Performance Trend")
```

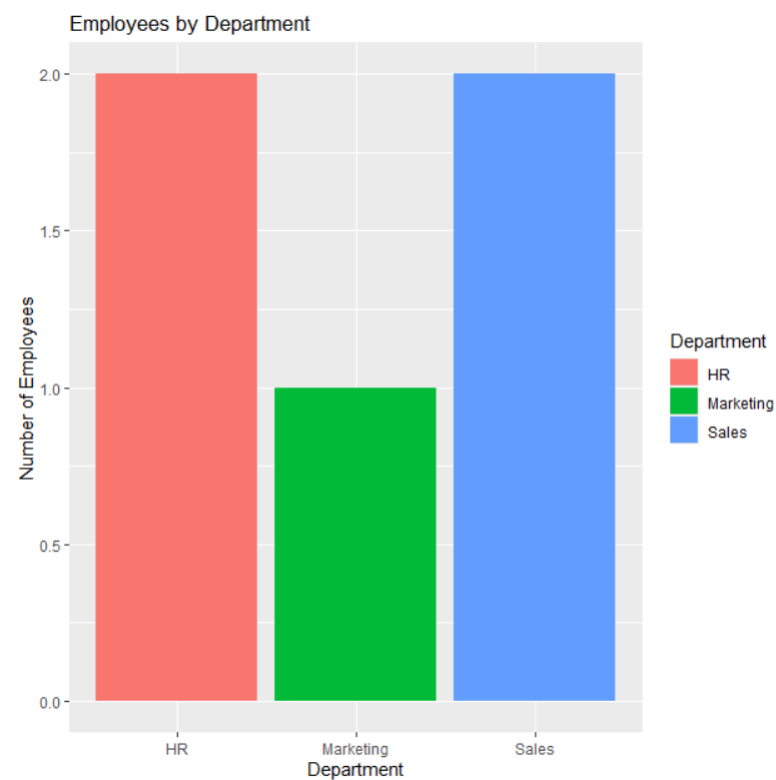
```
p2 <- ggplot(employee, aes(x = Department, fill = Department)) +  
  geom_bar() +  
  labs(title = "Employees by Department")
```

```
grid.arrange(p1, p2, ncol = 2)
```

1. Line Chart - Performance Trend



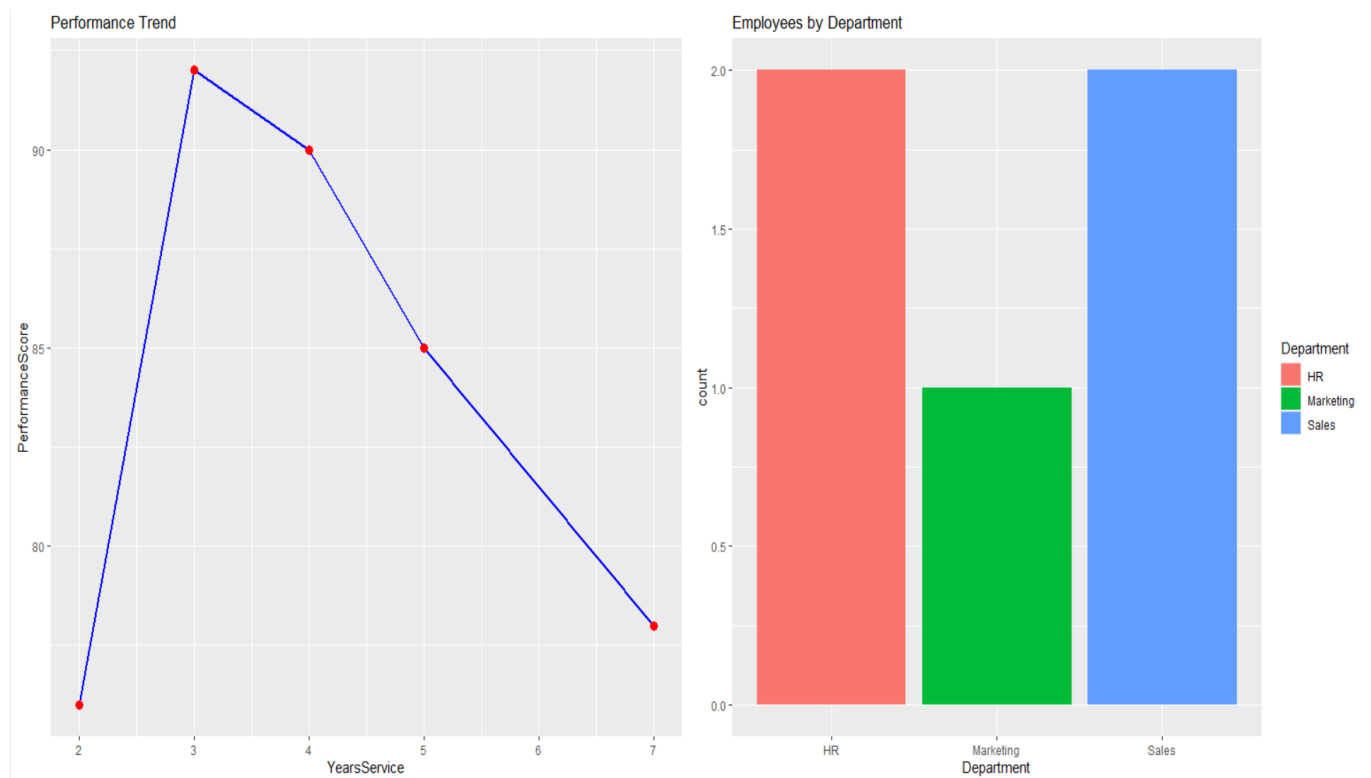
2. Bar Chart - Employees by Department



3. Scatter Plot - Years of Service vs Performance



4. Dashboard (Display Line Chart and Bar Chart Together)



Set 4: Product Inventory Management
Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.
4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Load libraries

```
library(ggplot2)
library(gridExtra)
```

Dataset

```
product <- data.frame(
  ProductID = c(1,2,3,4,5),
  ProductName = c("Product A","Product B","Product C","Product D","Product E"),
  Quantity = c(250,175,300,200,220),
  Category = c("Category 1","Category 1","Category 2","Category 2","Category 3"),
  Price = c(500,400,600,450,550)
)
```

1. Bar Chart

```
p1 <- ggplot(product, aes(x = ProductName, y = Quantity, fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(title = "Product Inventory",
       x = "Product Name",
       y = "Quantity Available")
```

```
print(p1)
```

2. Stacked Bar Chart

```
p2 <- ggplot(product, aes(x = Category, y = Quantity, fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(title = "Quantity by Product Category",
       x = "Category",
       y = "Quantity")
```

```
print(p2)
```

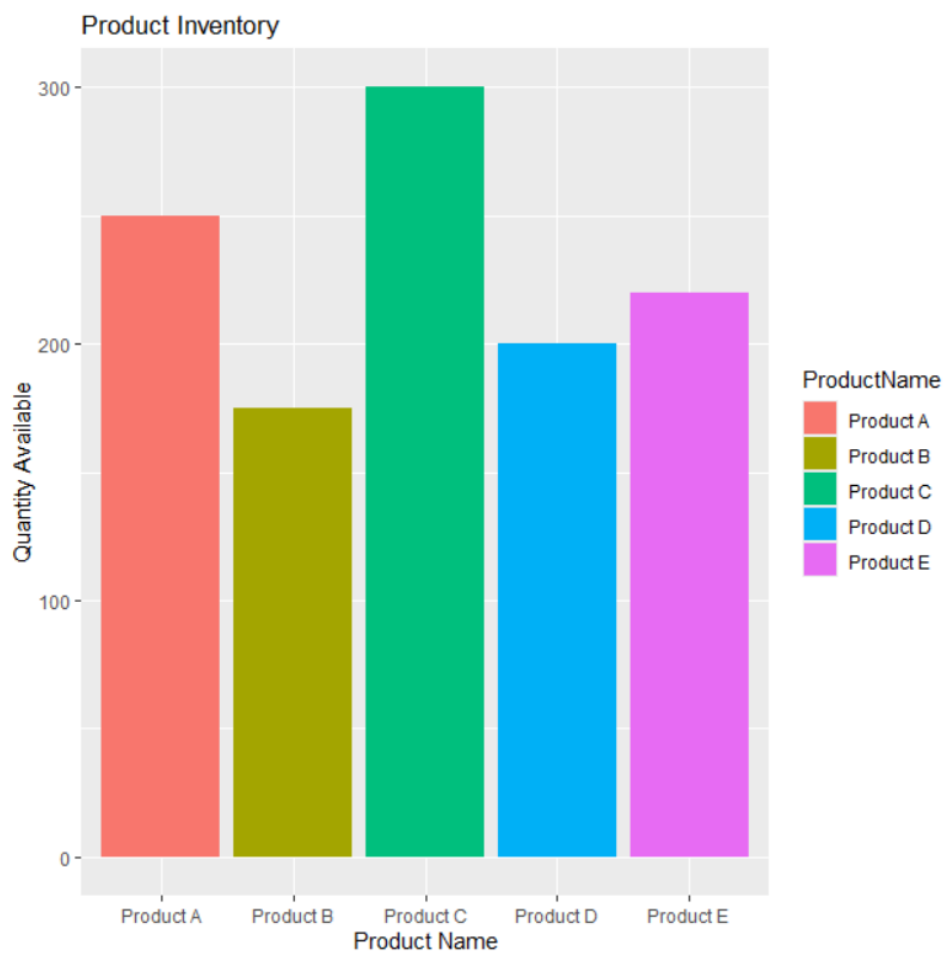
3. Scatter Plot

```
ggplot(product, aes(x = Price, y = Quantity, color = ProductName)) +  
  geom_point(size = 4) +  
  labs(title = "Price vs Quantity Available",  
        x = "Price",  
        y = "Quantity Available")
```

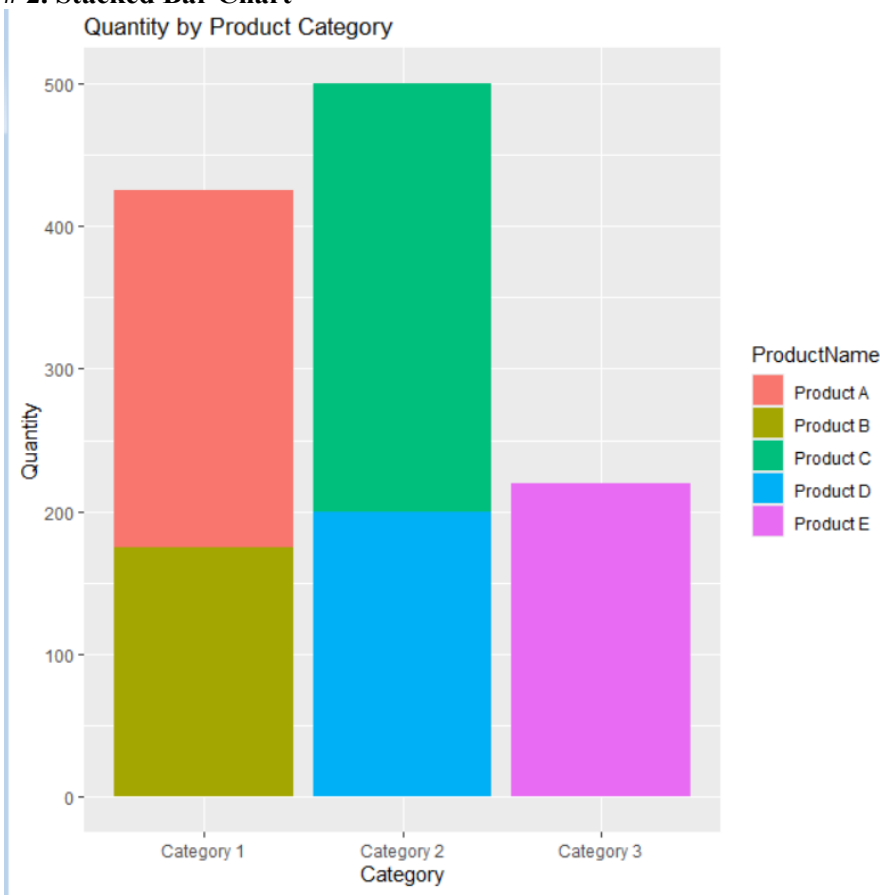
4. Dashboard

```
grid.arrange(p1, p2, ncol = 2)
```

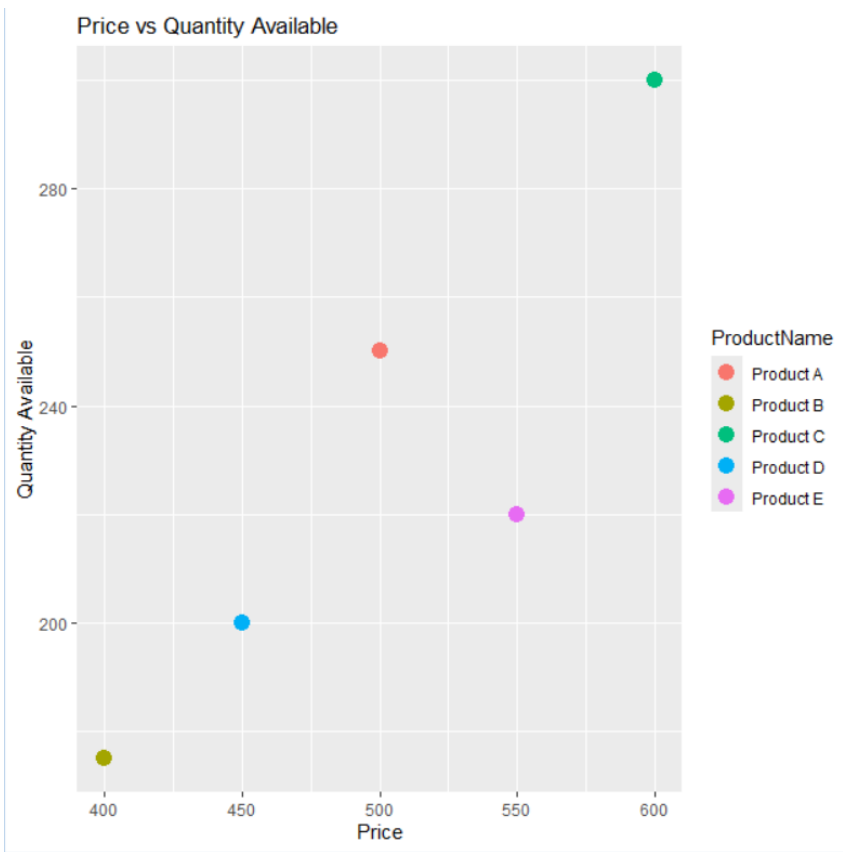
1. Bar Chart



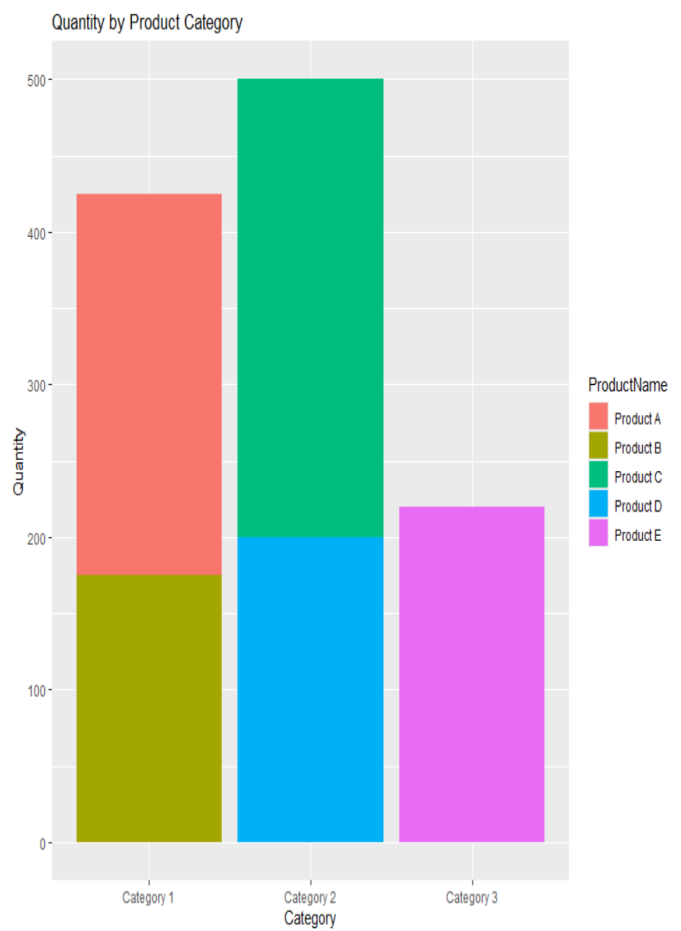
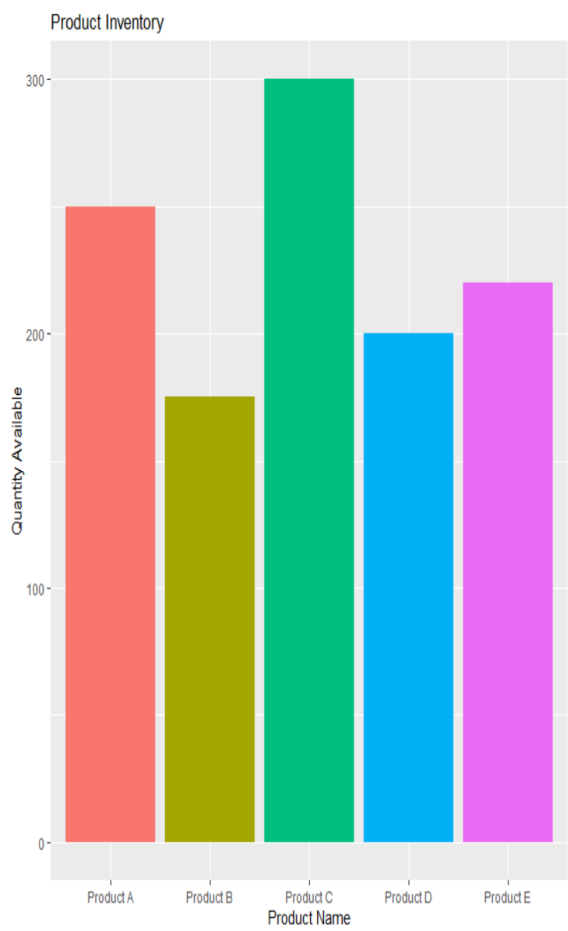
2. Stacked Bar Chart



3. Scatter Plot



4. Dashboard



Set 5: Website Analytics
Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

[Add comment](#)

2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.
4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

```
# Load libraries
```

```
library(ggplot2)
```

```
library(gridExtra)
```

Website Traffic Dataset

```
traffic <- data.frame(
```

```
  Date = c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05"),
```

```
  PageViews = c(1500,1600,1400,1650,1800),
```

```
  CTR = c(2.3,2.7,2.0,2.4,2.6),
```

```
  Likes = c(120,130,110,140,150),
```

```
  Shares = c(30,35,25,40,45),
```

```
  Comments = c(15,18,12,20,22))
```

1. Line Chart - Daily Page Views

```
p1 <- ggplot(traffic, aes(x = Date, y = PageViews, group = 1)) +
```

```
  geom_line(color = "blue", size = 1) +
```

```
  geom_point(color = "red", size = 3) +
```

```
  labs(
```

```
    title = "Daily Page Views",
```

```
x = "Date",  
y = "Page Views")
```

```
print(p1)
```

2. Bar Chart - Click Through Rate

```
p2 <- ggplot(traffic, aes(x = Date, y = CTR, fill = Date)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Click Through Rate",  
    x = "Date",  
    y = "CTR (%)")
```

```
print(p2)
```

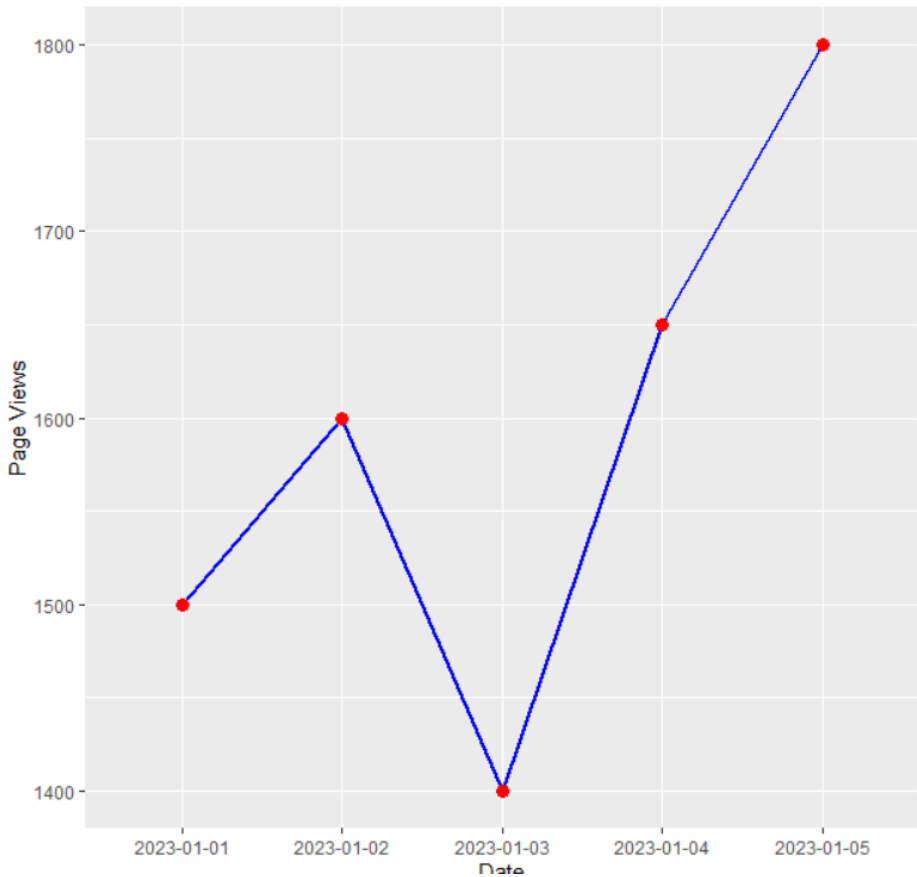
3. Stacked Area Chart - User Interactions

```
ggplot(traffic, aes(x = Date)) +  
  geom_area(aes(y = Likes, fill = "Likes"), alpha = 0.6) +  
  geom_area(aes(y = Shares, fill = "Shares"), alpha = 0.6) +  
  geom_area(aes(y = Comments, fill = "Comments"), alpha = 0.6) +  
  labs(  
    title = "User Interactions",  
    x = "Date",  
    y = "Count")
```

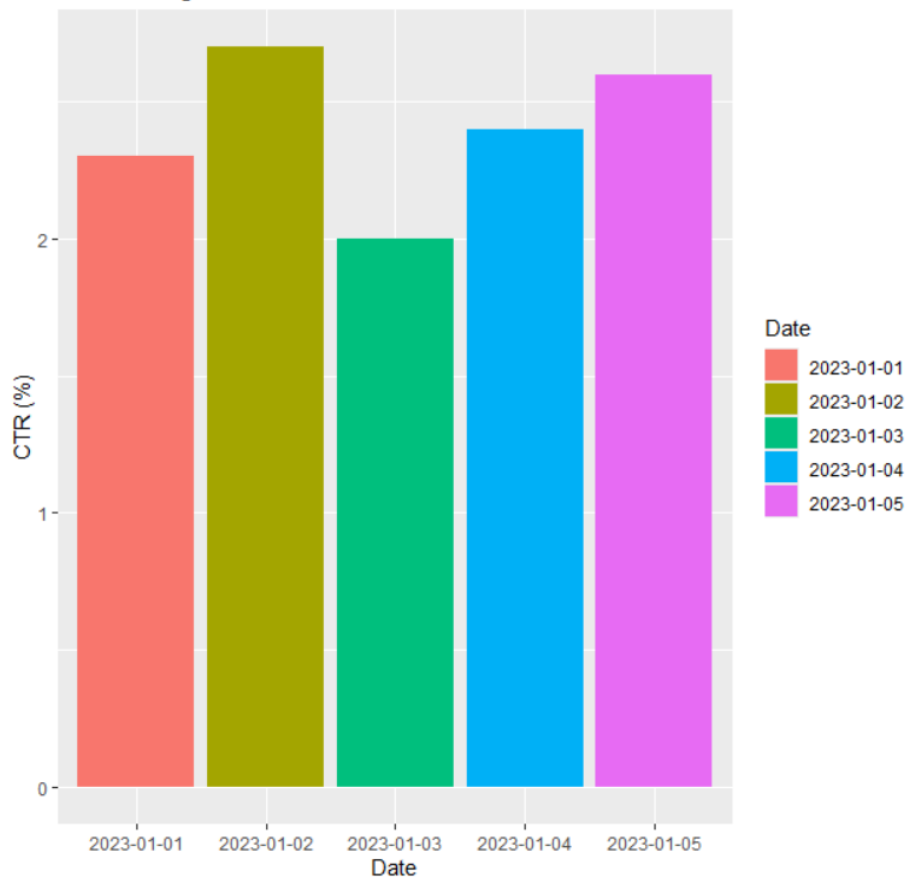
4. Dashboard

```
grid.arrange(p1, p2, ncol = 2)
```

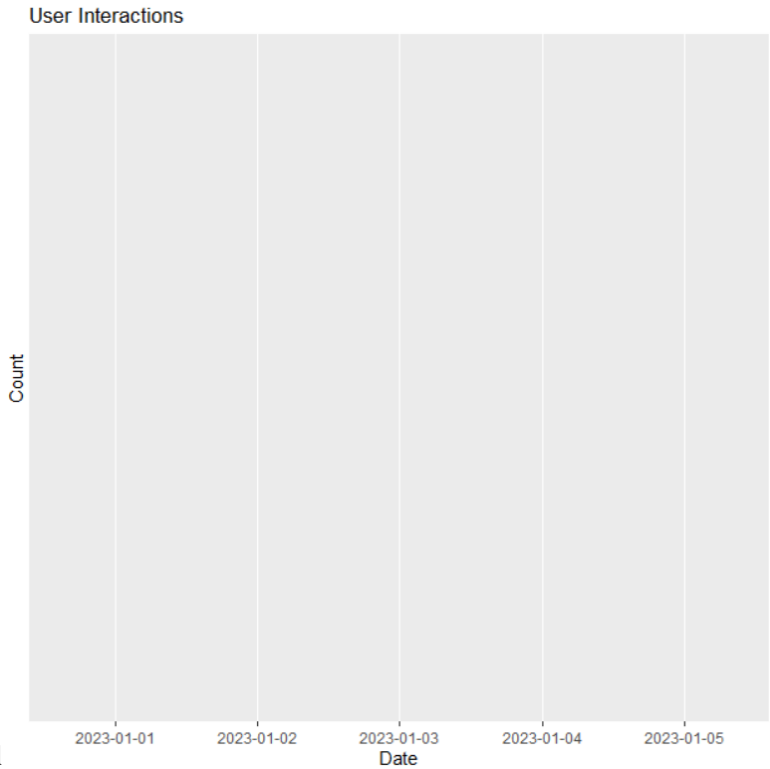
1. Line Chart - Daily Page Views
Daily Page Views



2. Bar Chart - Click Through Rate
Click Through Rate



3. Stacked Area Chart - User Interactions



4. Dashboard

